# Robed Beauvile-Owen, Ph.D.

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## EXPERIENCE

**Founder & Data Scientist** — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Location intelligence consultancy helping retail and service businesses decide where to expand. Turned owners' expansion questions into site-selection analyses; identified a $1.5M expansion opportunity.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, validated with difference-in-differences and a synthetic-data recovery test.
- Built low/base/high revenue projections and Monte Carlo financial risk models (probability of loss, VaR, CVaR) from client operating costs, local demand, and the causal estimates' 95% confidence intervals.
- Engineered Python ETL pipelines with automated data-quality checks and 150+ unit tests for Census, traffic, and competitor data, cutting market analysis turnaround by 91% (weeks to days).
- Designed a site-scoring model combining demographic, traffic, competitive-density, and consumer-behavior features, and explained results and uncertainty ranges to owners in written reports and maps.

**Senior Data Scientist** — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap with strategy, engineering, and behavioral science leadership, setting OKRs and translating business questions into models.
- Applied predictive, segmentation, and uplift models to millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible consumer records, cutting ad spend by $10K per campaign.
- Designed and standardized cross-team A/B testing and causal inference frameworks for direct-mail campaigns, applying CUPED to reduce metric variance by 20% and enable smaller tests.
- Owned the monthly model monitoring report covering drift, performance, and media mix forecast-vs-actual accuracy for up to 10 production models (media mix, response, eligibility, psychology-match).
- Scheduled Databricks jobs that ingested new data and refreshed model metrics, feeding the retraining and scoring pipelines behind every monitored model.
- Re-architected a production churn prediction platform in Python, SQL, Spark, Snowflake, and Databricks, reducing runtime by 48% and memory use by 23%.

**Founding Machine Learning Engineer** — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics, evaluation frameworks, and drift monitoring for production search and regression models, using offline validation and online experiments; mentored engineers and analysts on ML practices.
- Built SQL training tables that broke bulk orders into item-level shipping costs, then deployed a regression-based shipping-price API in C# on Azure, scaling to 10,000+ daily requests.
- Reduced customer dissatisfaction with search by 12% with product and engineering partners by building NLP relevance features: custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

**Data Scientist** — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built and validated XGBoost classification models on millions of healthcare consumer records with hundreds of external attributes, driving a 3% lift in response rates; explained model drivers to stakeholders with SHAP.
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, removing 60 minutes of manual work daily.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas, focusing campaign resources on high-opportunity regions.

**Graduate Researcher & Teaching Assistant** — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in mathematical optimization, functional analysis, and operator theory for doctoral dissertation research.
- Taught mathematics and statistics to classes of 75-200 students and mentored 100+ students in summer programs, achieving a 95% pass rate.

## TECHNICAL PROJECTS

**CaLaun** — *AI-Powered Calculus Tutor (LLM Application)* github.com/calaun-learning/calaun

- Built and deployed a generative AI tutor that grounds a self-hosted Llama 3.2 LLM in SymPy-verified solution steps, scoping each prompt to one step so the model explains but never computes; 400+ weekly users, 90+ automated tests.

**Medicare Claims Intelligence** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a tested dbt warehouse over 790K CMS Medicare claims and a provider fraud (FWA) detector that flagged planted billing schemes with 81% precision.
- Cut forecast error 41% vs. a seasonal benchmark (ARIMA, ETS, Prophet); built a SHAP-explained XGBoost model that finds 46% more high-cost members than baseline (MLflow) and a RAG assistant with PHI guardrails.

**Customer LTV & Churn Survival Modeling** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes to show where LTV = ARPU / churn fails, estimating customer lifetime value from empirical survival curves.

## EDUCATION

**Ph.D. in Mathematics** — *University of Pittsburgh* 2017 - Dec 2023

**Dissertation:** A Minimal Invariant Set, Frames, and Operators

**B.S. in Mathematics** — *Florida State University* 2014 - 2017

**Minor:** Philosophy | **GPA:** 3.90

## TECHNICAL SKILLS

**Programming:** Python (pandas, NumPy, scikit-learn, statsmodels, PyTorch, GeoPandas), SQL, C#, Bash, PowerShell

**Machine Learning & AI:** Regression, Classification, Clustering, XGBoost, Anomaly Detection, Time Series Forecasting (ARIMA, ETS, Prophet), Survival Analysis, Uplift Modeling, Media Mix Modeling, SHAP, NLP, LLMs (Llama), RAG, Prompt Engineering, Feature Engineering, Model Monitoring & Drift

**Statistics & Experimentation:** Hypothesis Testing, A/B Testing, CUPED, Power Analysis, Causal Inference, Double ML, Causal Forests, Difference-in-Differences, CATE, Monte Carlo Simulation, Optimization

**Data Engineering & Cloud:** Spark, Databricks (Workflows), Snowflake, dbt, MLflow, AWS, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT Pipelines, Data Quality Checks, Unit Testing, Git, AI-Assisted Development (Claude Code)

**Visualization & Communication:** Power BI, Tableau, Excel, Matplotlib, Plotly, Dashboards, Stakeholder Presentations, Mentoring